

Quantum Computing & Cryptographic Risk

What quantum does (and doesn't) break

Shor's algorithm (theoretical): breaks RSA, ECC (factoring/discrete log) efficiently

Grover's algorithm: halves symmetric key strength (AES-256 → ~128 effective)

Quantum machines today: small-scale; breaking RSA-2048 is not yet practical, but the threat is real enough to act

What is NOT broken

Symmetric crypto (AES-256 with minor param bump)

Hashing (SHA-256 is fine)

Post-quantum schemes (NIST PQC: Kyber, Dilithium, etc.)

NIST standards (the migration guide)

Kyber (ML-KEM) → key exchange (used in TLS)

Dilithium (ML-DSA) → digital signatures (certs, code-sign)

Why 'harvest now, decrypt later' matters

An adversary captures encrypted data TODAY, decrypts it in 10 years when QC is viable

Long-lived secrets (patient data, state secrets, key material) need quantum-resistant protection NOW

TLS already supports PQC in some libraries (Google Chrome, Cloudflare) – the migration has begun

Migration strategy

Inventory: where is RSA/ECC used in your stack (keys, certs, protocols)

Classify by data lifetime: short (1yr) vs long (10yr+)

Start: rotate to PQC key exchange in TLS (Kyber)

Plan: migrate code-signing and long-lived keys to Dilithium

Test: PQC libraries have different sizes/perf – measure

The realistic view

Doomsday QC is not imminent; NIST says plan now

Governments already mandate PQC migration timelines

The risk is real for long-lived secrets; 'wait and see' is a liability if data lifetime > time-to-QC

The blue team takeaway

Know your crypto inventory. The org that doesn't know where RSA lives will scramble when the standard rolls out.

Quantum computing doesn't 'break everything' – it breaks RSA/ECC. Act now on long-lived secrets; plan migration for the rest; symmetric crypto stays safe.

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